

Computer Science & IT

Discrete Mathematics



Graph Theory

Lecture No. 11



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Recap of Previous Lecture



- 
- ★ Topic Maximal matching
 - ★ Topic Maximum matching
 - ★ Topic Matching number
 - ★ Topic Perfect matching

Topics to be Covered



Topic

Line / Edge Covering



Topic

Minimal & Minimum line covering



Topic

Line covering number



Topic

Line independent set



Topic

Maximal & Maximum line independent set



Line independence number

Covering

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graph TD; Covering --> LineCovering[Line Covering / Edge Covering]; Covering --> VertexCovering[Vertex Covering];
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Line Covering / Edge Covering

We will try to cover all the vertices of the graph using subset of "edges/lines"

Vertex Covering

We will try to cover all the edges of the graph using subset of "vertices"



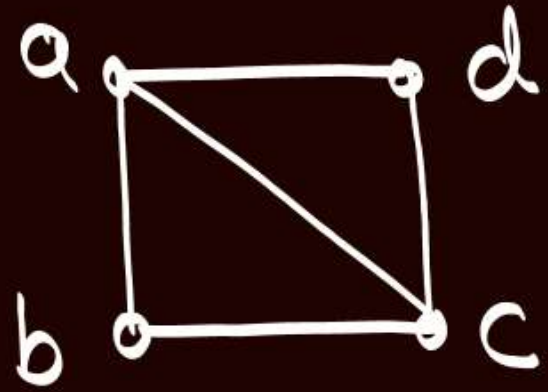
Topic : Line/Edge Covering

★ Let $G=(V,E)$ be a graph.

A subset C of set of edges E is called line covering of graph G if every vertex of the graph is incident with at least one edge of subset C .

i.e, in a subgraph corresponding to the line covering
 $\deg(v) \geq 1, \forall v \in G$

eg:-



"G"

$V = \{a, b, c, d\}$

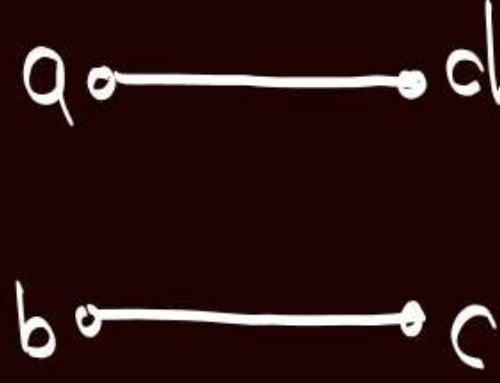
$E = \{\{a, b\}, \{a, c\}, \{a, d\}, \{b, c\}, \{b, d\}\}$

$C_1 = \{\{a, b\}, \{c, d\}\}$

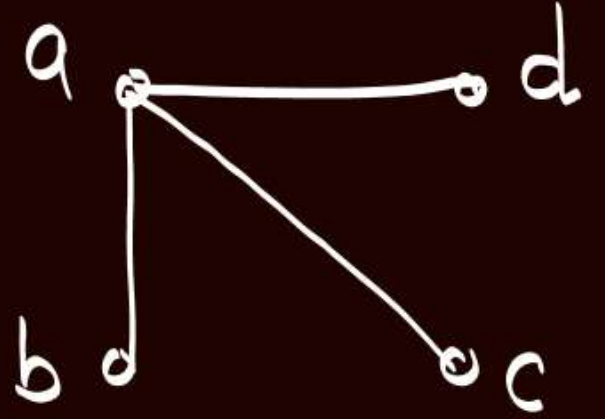


"C₁"

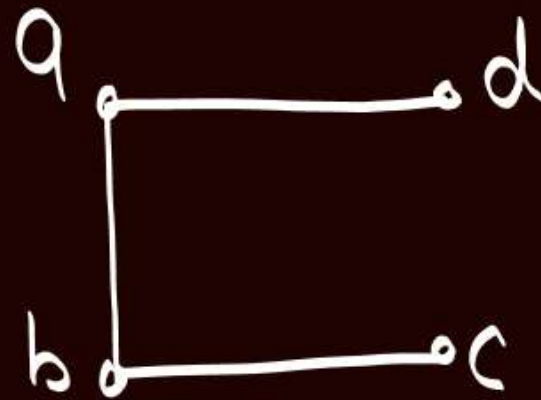
$C_2 = \{\{a, d\}, \{b, c\}\}$



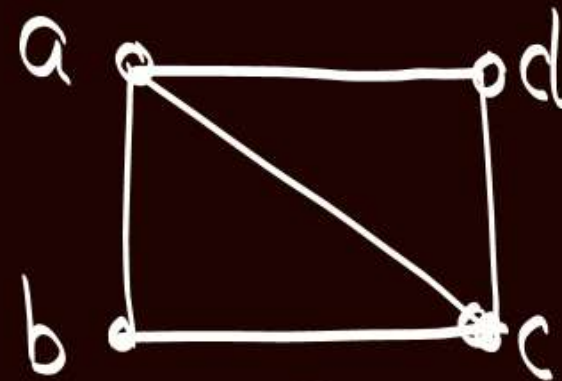
"C₂"



C₃



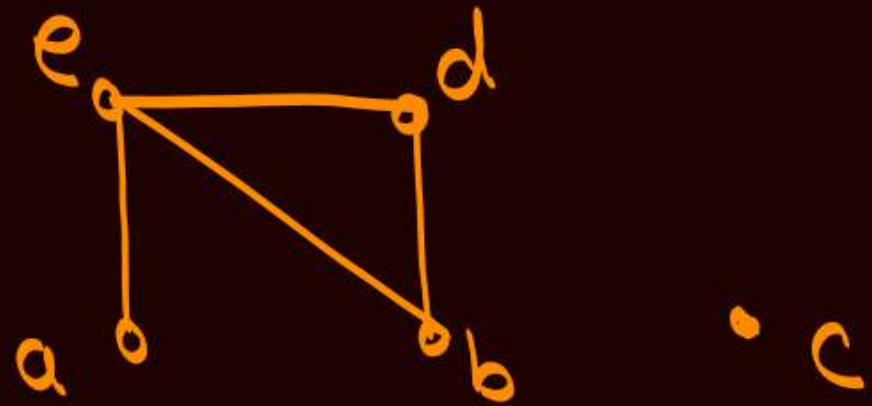
C₄



C₅ = E

C₁, C₂, C₃, C₄, C₅
etc. are line
covering of
graph G

→ Consider the following graph G



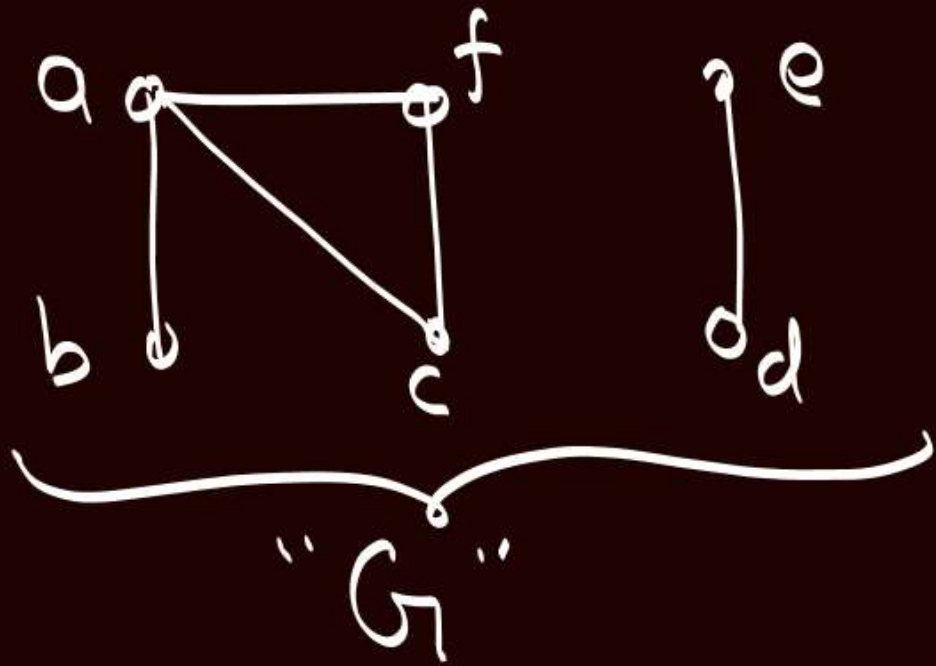
do we have any line covering for graph G .

Ans: No line covering can exist for the given graph

Line covering does not exist for graph G
if and only if

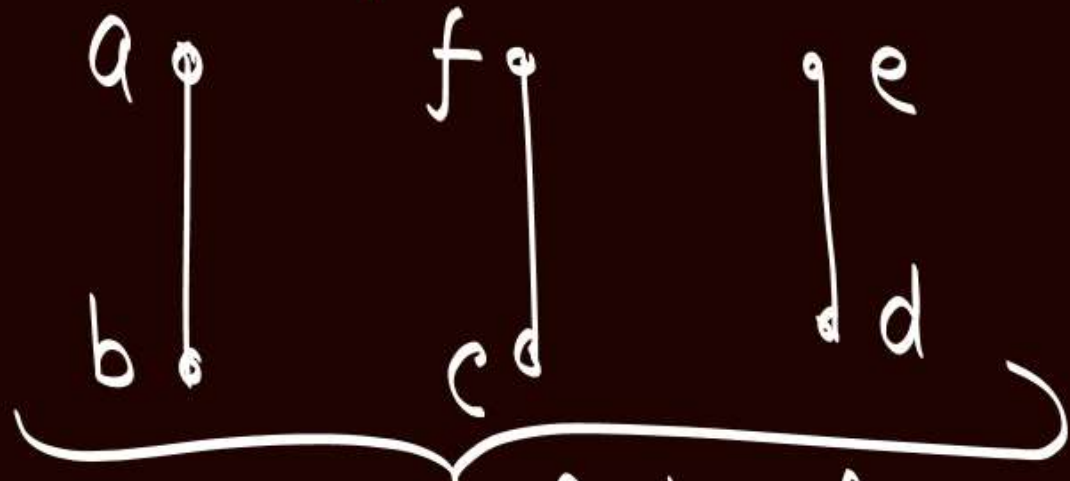
at least one isolated vertex exist in graph G

eg:



Does line covering exist for graph G .

Ans = yes.



is one of the line covering
for graph G

Line covering may exist
for a disconnected graph
as well.
{ Provided No isolated
Vertex exist in the graph }



Topic : Minimal line covering

A line covering of graph G is said to be minimal if no edge can be deleted from that line covering without destroying its property to cover all the vertices of the graph.

In the above example,

C_1 , C_2 and C_3 are minimal line coverings of graph G .



Topic : Minimum line covering

Smallest minimal line covering



- Line coverings of graph G with minimum number of edges are called minimum line coverings of graph G .

In the above example,

C_1 and C_2 are minimum line covering of graph G .

Note:-

- ① Every minimum line covering of graph G is also a minimal line covering, but every minimal line covering need not be a minimum line covering.
- ② A minimal line covering of graph G can never contain any cycle.
- ③ In a minimal line covering of graph G there exist no path of length '3' or 'more'.



Topic : Line covering number (α_1)

Line covering no. of graph G = No. of edges in any one of the minimum line covering

α_1 = No. of edges in any one of the minimum line covering

In the above example,
line covering No. of $G = 2$
 $\{w.r.t. C_1 \text{ or } C_2\}$

For any graph G with n -vertices
line covering No. of $G \geq \lceil \frac{n}{2} \rceil$



Topic : Line independent set

Let $G = (V, E)$ be a graph.

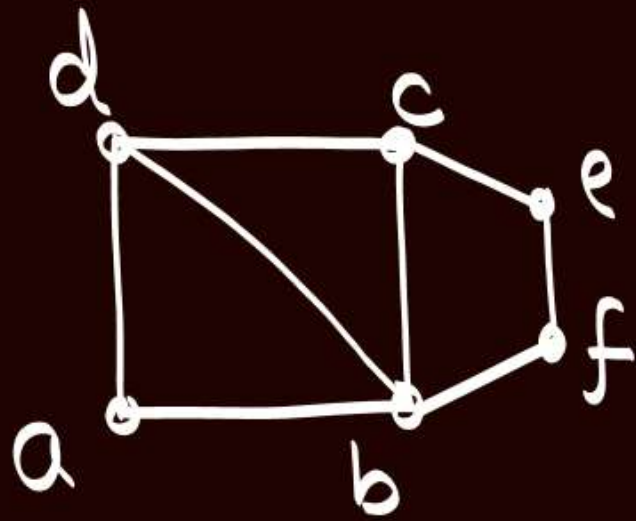
A subset 'L' of set of edges 'E' is called a line independent set if no two edges of subset L are adjacent to each other.

No two edges can be adjacent to each other

∴ in a subgraph w.r.t. line independent set, $\deg(v) \leq 1, \forall v \in G$

Note: $\deg(v) \leq 1, \forall v \in G$ is also the condition for Matching of graph G.

Hence Every line independent set of graph G is also a matching of graph G and vice-versa



G

$$L_1 = \{\{a, d\}\}$$

{ A set containing a single edge }
is always a line independent set }



" L_1 " is a matching as well

$$L_2 = \{\{a, d\}, \{b, c\}\}$$

$$L_3 = \{\{b, d\}, \{e, f\}\}$$

$$L_4 = \{\{a, d\}, \{b, c\}, \{e, f\}\}$$

$$L_5 = \{\{a, b\}, \{d, c\}, \{e, f\}\}$$

L_1, L_2, L_3, L_4, L_5 etc.
are line independent set
w.r.t. graph G .

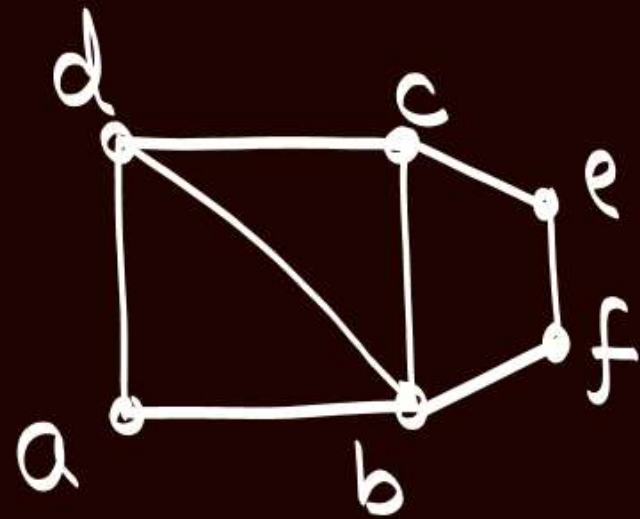


Topic : Maximal Line independent set

- * A Line independent set is called a maximal line independent set if no new edge from the graph can be added to the set without destroying its property of being independent set.

In the above example,

L_3, L_4 & L_5 are maximal line independent set w.r.t. graph G .



G

Not maximal

$$L_1 = \{\{a, d\}\}$$

{ A set containing a single edge }
is always a line independent set }



" L_1 " is a matching as well

$$L_2 = \{\{a, d\}, \{b, c\}\}$$

$$L_3 = \{\{b, d\}, \{e, f\}\}$$

$$L_4 = \{\{a, d\}, \{b, c\}, \{e, f\}\}$$

$$L_5 = \{\{a, b\}, \{d, c\}, \{e, f\}\}$$

Maximal

L_1, L_2, L_3, L_4, L_5 etc.
are line independent set
w.r.t. graph G .

Maximum
as well



Topic : Maximum Line independent set

Largest Maximal
Line independent Set



- * Line independent sets with maximum number of edges are called maximum line independent sets.

In the above example,

L_4 & L_5 are maximum line independent set w.r.t. graph G .



Subgraphs w.r.t. maximum line independent set
will define "maximum matching" for the graph.

→ Every maximum line independent set is also a maximal line independent set, but Every maximal line independent set need not be a maximum line independent set.



Topic : Line independence number (β_1)

Line independence Number of graph G = Number of edges in any one of the maximum line independent set of G .

β_1 = No. of edges in any one of the maximum line independent set = Matching No. of graph G

For a graph G with ' n ' vertices

$$\text{Line independence no. of graph } G \leq \left\lfloor \frac{n}{2} \right\rfloor$$

In the above example,

Line independence no. of $G(P_1) = 3$

w.r.t. L_4 or L_5

* For any graph $G = (V, E)$

$$\alpha_1 + \beta_1 = |V(G)|$$

i.e. (Line covering number of graph G + Line independence number of graph G) = Total number of vertices in graph G

Q. Find α_1 & β_1 for Complete graph K_n .

Line covering no. (α_1) for $K_n = \left\lceil \frac{n}{2} \right\rceil$

→ Line independence no. (β_1) for $K_n = \left\lfloor \frac{n}{2} \right\rfloor$

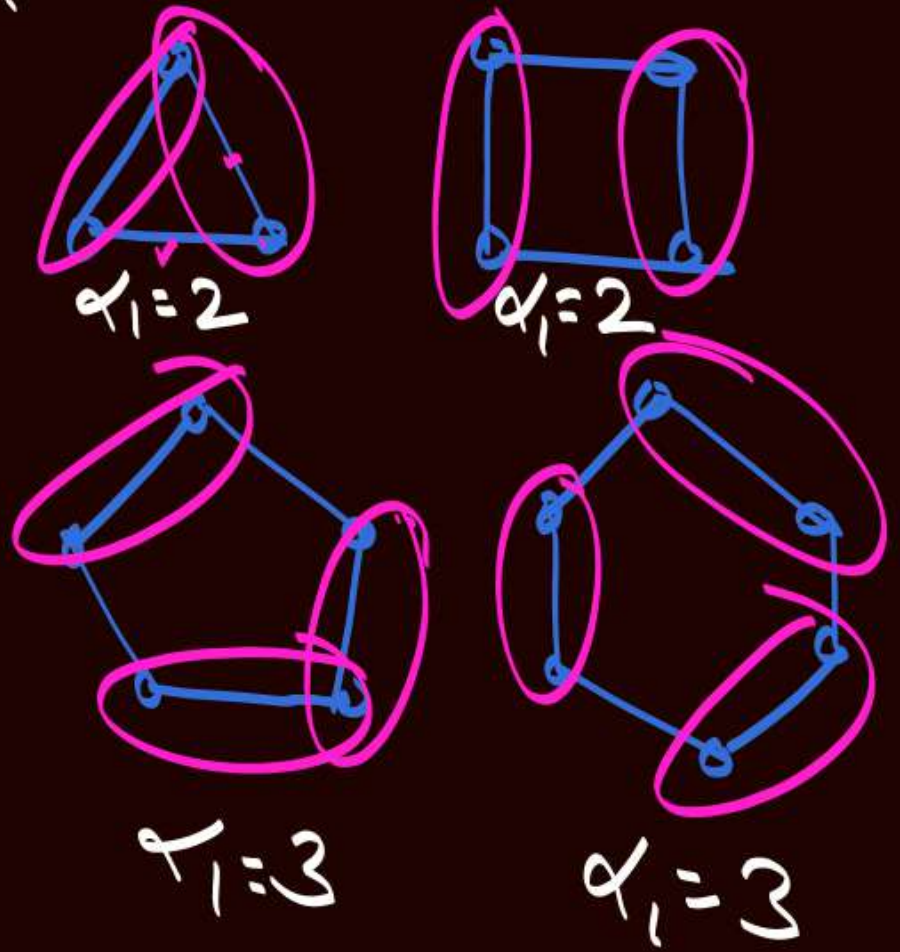
We know, $\alpha_1 + \beta_1 = n$

Q. Find α_1 & β_1 for Cycle graph C_n

Line covering no. (α_1) for $C_n = \left\lceil \frac{n}{2} \right\rceil$

Line independence no. (β_1) for $C_n = \left\lfloor \frac{n}{2} \right\rfloor$

We know, $\alpha_1 + \beta_1 = n$

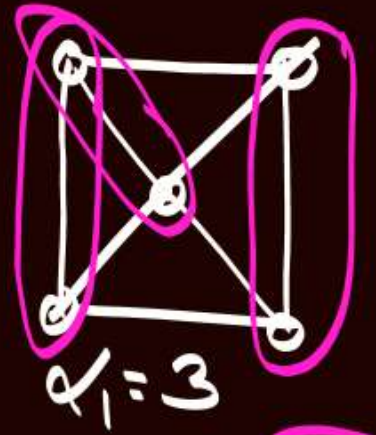


Q. Find α_1 & β_1 for Wheel graph W_n

Line covering no. (α_1) for $W_n = \left\lceil \frac{n}{2} \right\rceil$

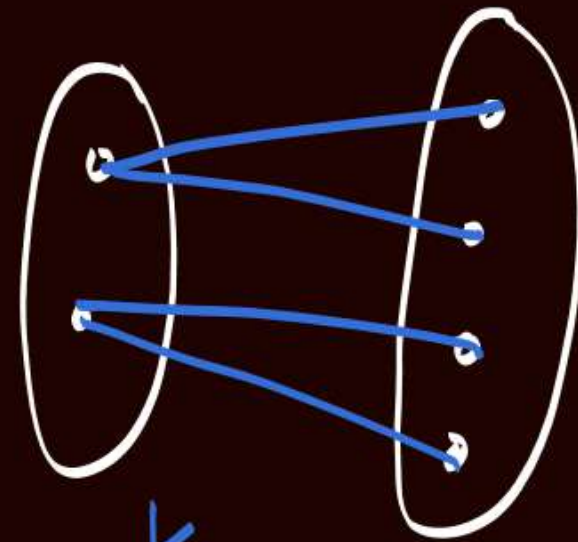
Line independence no. (β_1) for $W_n = \left\lfloor \frac{n}{2} \right\rfloor$

We know $\alpha_1 + \beta_1 = n$



Q. Find α_1 & β_1 for Complete bipartite graph $K_{m,n}$

Line covering no. (α_1) for $K_{m,n} = \max(m,n)$

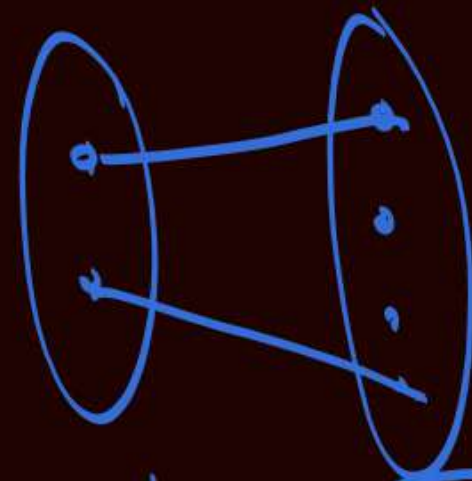


$K_{2,4}$

$\alpha_1 = 4$

Line independence no. (β_1) for $K_{m,n} = \min(m,n)$

We know, $\alpha_1 + \beta_1 = \underbrace{(m+n)}_{\text{Total no. of vertices}}$



$K_{2,4}$

$\beta_1 = 2$

Q. Find α_1 & β_1 for Star graph with n -vertices.

Line covering no. (α_1) for
Star graph with n -vertices
 $\equiv K_{1,n-1}$ or $K_{n-1,1}$

$$= \max(1, n-1) = (n-1)$$



Line independence no. (β_1) for
Star graph with n -vertices
 $\equiv K_{1,n-1}$ or $K_{n-1,1}$

$$= \min(1, n-1) = 1$$



2 mins Summary



✓
Topic

Line / Edge Covering

✓
Topic

Minimal & Minimum line covering

✓
Topic

Line covering number

✓
Topic

Line independent set

✓
Topic

Maximal & Maximum line independent set

THANK - YOU